QUASI-AFFINE FINITE TYPE MORPHISMS SIT CLOSED INSIDE A FINITE PRESENTATION ONE

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 3019-3033, following the statement at 3007-3017.

Editorial note. No omissions.

Allowed standard prerequisites (editorial).

- Characterization of quasi-affine morphisms. Morphisms of Spaces, Lemma spaces-morphisms-lemma-characterize-quasi-affine (cross-chapter, cited at ~3019-3021). Factorization X -> Z -> Y with X -> Z quasi-compact open immersion and Z -> Y affine.
- Affine morphisms are limits of fp affines. lemma-affine-morphism-is-limit (same chapter, cited at ~3022-3023). Z = lim Z_i with Z_i affine fp over Y.
- Opens descend to finite stages and eventual closedness. lemma-descend-opens (same chapter, cited at ~3024-3026); lemma-finite-type-eventually-closed (same chapter, cited at ~3029-3031). X is the inverse image of a quasi-compact open U_0 in Z_0; X -> U_i is a closed immersion for i large.

Source reference. Original lemma 0870: sources/stacks-spaces-limits/pinned-original-source.tex, source lines 2917–2917. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 0827: sources/stacks-spaces-limits/pinned-original-source.tex, source lines 1295–1295. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 0828: sources/stacks-spaces-limits/pinned-original-source.tex, source lines 1452–1452. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 086S: focused-seeds/geometry/S9-spaces-limits-quasi-affine-closed-in-finite-presentation/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input;

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the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let S be a scheme. Let $f : X \to Y$ be a morphism of algebraic spaces over S. Assume*

(1) *f is locally of finite type and quasi-affine, and*
(2) *Y is quasi-compact and quasi-separated.*

Then there exists a morphism of finite presentation $f' : X' \to Y$ and a closed immersion $X \to X'$ over Y.

Proof. By Morphisms of Spaces, Lemma 086S we can find a factorization $X \to Z \to Y$ where $X \to Z$ is a quasi-compact open immersion and $Z \to Y$ is affine. Write $Z = \lim Z_i$ with Z_i affine and of finite presentation over Y (Lemma 0870). For some $0 \in I$ we can find a quasi-compact open $U_0 \subset Z_0$ such that X is isomorphic to the inverse image of U_0 in Z (Lemma 0827). Let U_i be the inverse image of U_0 in Z_i, so $U = \lim U_i$. By Lemma 0828 we see that $X \to U_i$ is a closed immersion for some i large enough. Setting $X' = U_i$ finishes the proof. □